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MILITARY SPECIFICATION

TEST SET, ORGANIZATIONAL MAINTENANCE AN/TPM-23

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Organizational Maintenance Test Set AN/TPM-23 (hereinafter referenced as OMTS) a portable test set used for second echelon (organizational) maintenance of Radar Set AN/VPS-2. The OMTS contains test equipment to locate defective units, assemblies, or subassemblies of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel: Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction; M18; Manufacturing Process and Inspection of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Frankford Arsenal, ATTN: SARFA-MDM, Phila., Pa. 19137, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4931

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	Environmental Test Methods

DRAWINGS

U. S. Army, Frankford Arsenal

D10548613	7.5 MHz Clock Generator
F10554522	Test Set, Organizational Maintenance AN/TPM-23
F10549202	Transmit Gate
D10548158	60MHz Local Oscillator
D10548183	Hybrid Junction
D10556252	Connector, Plug, Electrical
C10558457	Variable Attenuator

PACKAGING DATA SHEET

10554522	Packaging of Test Set, Organizational Maintenance AN/TPM-23

(Copies of specification, standards, drawings, and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The OMTS shall be manufactured in accordance with Drawing F10554522 and drawings pertaining thereto.

3.2 General specification.- Where applicable, the OMTS shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

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3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishings
d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

3.2.2 MIL-C-46357

a. Etched circuit assemblies
b. Wiring
c. Threaded fasteners
d. Marking
e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration.- The OMTS shall be mounted on the vibration table in normal service position. The OMTS shall meet the requirements of 3.4 and 3.5 after being subject to continuous vibration for two hours along each of three mutually perpendicular axes within the frequency range and amplitude as follows:

Frequency Range	Double Amplitude
5 to 15	0.06 inch
15 to 25	0.04 inch
25 to 55	0.02 inch

The rate of change of frequency at each frequency range shall not exceed one cycle-per-second per second (1 cps/sec). Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock.- The OMTS shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (g's) and a pulse duration of 11 ± 1 milliseconds.

3.3.3 Altitude.

3.3.3.1 Operating.- The OMTS shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

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3.3.3.2 Nonoperating.- The OMTS shall meet the requirements of 3.4 and 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating.- The OMTS shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40°F and +130°F.

3.3.4.2 Nonoperating (storage).- The OMTS shall not suffer permanent damage when subjected to the temperature extremes of -80°F, and +185°F, and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity-nonoperating.- The OMTS shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days with the combination case closed and latched.

3.3.6 Drop test.- The OMTS combination case with the cover closed and latched containing a dummy load of the size, weight and shape of the equipment shall be subjected to a total of 14 drops, once on each corner and each face from a height of 24 inches. Subsequent to these drops, there shall be no evidence of damage such as openings along joint, loosening of partitions, splitting, cracking of material, partial or total opening of latches. The case and cover shall function in the same manner as prior to the drop test.

3.3.7 Pressure retention.- After the OMTS combination case has been subjected to the drop test, and the case closed and latched, the case shall maintain a positive pressure of 1 psi for not less than 5 minutes. A loss of pressure in excess of .1 psi will be considered a failure.

3.3.8 Electromagnetic interference.- The OMTS shall meet the requirements of MIL-STD-461 for type IC equipment.

3.3.9 Sand and dust.- The OMTS shall withstand 3 hours of continuous exposure to blowing sand of particle size 0.01 to 1.00mm and blowing dust of particle size 0.001 to 0.01mm with a density between 0.1 to 0.5 grams per cubic foot of sand and a density between 0.1 to 0.5 grams per cubic foot of dust at a temperature of plus 75°F, a relative humidity between 15 to 30%, and a minimum velocity of 15 knots. Subsequent to sand and dust exposure, the OMTS shall show no evedence of damage or be adversely affected and shall meet all requirements of this specification. This test shall be performed with the case closed and latched.

3.3.10 Salt spray.- The OMTS shall withstand 48 hours of continuous exposure to a fine spray of 5% salt solution concentration with a specific gravity of from 1.0268 to 1.0413, a pH between 6.5 and 7.2, and a temperature between 93° and 97°F. Subsequent to salt spray, the OMTS shall show

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no evidence of damage or be adversely affected, and shall meet all the requirements of this specification. This test shall be performed with the case closed and latched.

3.3.11 Fungus.- The OMTS shall be capable of 28 days exposure to a composite spore suspension consisting of the following cultures:

(a) Chatomium globosum

(b) Aspergillus versicalor

(c) Aspergillus flavus

(d) Pencillium luteum

(e) Rhizopus nigricans

The composite spore suspension shall be incubated at an ambient temperature of between +77°F and +84°F with an RH of 97 ±2%. Subsequent to the above fungus exposure, the OMTS shall meet all the requirements of this specification. This test shall be performed with the case closed and latched.

3.4 Electrical.

3.4.1 Continuity.- Each conductor shall be continuous electrically as specified on the schematic diagram.

3.4.2 Resistance.

3.4.2.1 Antenna test section.- Resistance between the following terminal points shall be as follows for the temperatures specified:

Terminal Points	60° to 90°F (Ambient temperature) Resistance (ohms)	-40° and +130°F (Extreme Operating Temperatures) Resistance (ohms)
S1 (TP19) to S2 (TP20)	170 to 230	140 to 240
S2 (TP20) to S3 (TP21)	170 to 230	140 to 240
S3 (TP21) to S1 (TP19)	170 to 230	140 to 240
R1 (TP22) to R2 (TP23)	350 to 450	320 to 490
R2 (TP23) to R3 (TP24)	350 to 450	320 to 490
R3 (TP24) to R1 (TP22)	350 to 450	320 to 490

3.4.2.2 High voltage load section.- With +24 volts dc applied to PWR jack (J6) resistances between KLYSTRON jack (J7), the high voltage interlock connector and the OMTS chassis shall be as follows:

(a) With 2A1A3/2A6 switch S6 set to 2A1A3.

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Terminals	Resistor (ohms)
High voltage interlock connector to chassis	greater than 10 meg
High voltage interlock connector to J7-A	0.5 meg +20% -15%
High voltage interlock connector to J7-B	1.0 meg +20% -15%
J7-A to chassis	not less than 10 meg
J7-A to J7-B	0.5 meg +20% -15%
J7-B to chassis	not less than 10 meg
J7-C to chassis	100 k ±1%

(b) With 2A1A3/2A6 switch S6 set to 2A6:

Terminals	Resistance (ohms)
J7-C to J7-B	less than 1.0
High voltage interlock connector to chassis	1.1 meg +20% -15%

3.4.2.3 Range test section.- Resistance between EXT MTR HIGH tip jack (TP16) and R RTN tip jack (TP17) for each position of TEST SELECTOR switch (S7) shall be as follows:

TEST SELECTOR position	Resistance in ohms	
OFF	17,100	18,900
1	15,700	17,400
2	15,700	17,400
3	15,700	17,400
4	17,100	18,900
5	17,100	18,900
6	17,100	18,900
7	17,100	18,900
8	17,100	18,900

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	9	less than 1.0

3.4.3 Power switch and indicator lamp.- With +24 +6 -2 volts dc power connected to 24V PWR jac, (J6) and 24V PWR circuit breaker (CB1) energized, POWER indicator lamp (DS1) shall illuminate when POWER switch (S5) is set to ON. The lamp shall extinguish when S5 is set to OFF.

3.4.4 Power circuit breaker.- The 24V PWR circuit breaker (CB1) shall be energized when reset pushbutton is depressed so white band is not exposed. The circuit breaker shall be deenergized when reset pushbutton is pulled out (manually or by overload) to expose white band.

3.4.5 Dc power regulation.- With 24V PWR circuit breaker (CB1) energized and power switch (S5) set to ON, voltage across zener diode VR1 shall be 4.85 to 5.35 volts when input power applied to 24V PWR jack (J6) is varied between +22 and +30 volts dc.

3.5 Performance.

3.5.1 Antenna test section.

3.5.1.1 EL/TRAV POS synchro differential transmitter alignment.- EL/TRAV POS synchro differential transmitter (B1) shall be locked at alignment of electrical zero.

3.5.1.2 EL/TRAV POS synchro differential transmitter voltages.- With an external synchro control transmitter aligned and locked at electrical zero and excited by 115 volts, 400 Hz, connected to the stator of EL/TRAV POS synchro differential transmitter (B1) of the OMTS, ac voltages across rotor leads R1, R2, and R3 at specified angles of rotation of EL/TRAV POS dial shall be as follows:

	Angle of Rotation (degrees)	Voltage
R1-R3	0 ± 5	less than 1
	90 ± 5	78 ±7
	180 ± 5	less than 1
	270 ± 5	78 ±7
R1-R2	60 ± 5	less than 1
	150 ± 5	78 ±7
	240 ± 5	less than 1

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	330 ± 5	78 ± 7
R2-R3	30 ± 5	78 ± 7
	120 ± 5	less than 1
	210 ± 5	78 ± 7
	300 ± 5	less than 1

3.5.2 Signal test section.

3.5.2.1 Signal selector switch and meter.- The SIGNAL SELECTOR switch (S4) shall be set in turn to positions 1 through 9, with the appropriate test input singal applied for each position. The proper meter indication for each switch position shall be observed on SIGNAL INDICATOR meter (M1).

(a) Position 1.- With an RF signal having the following characteristics connected to RF INPUT jack (J4), meter shall read between 5 and 7.5 milliamperes:

Type - CW (unmodulated)

Frequency - 60 ±1 MHz

Average power - +23 dBm

(b) Position 2.- With an RF signal having the following characteristics connected to RF INPUT jack (J4), meter shall read between 5 and 7.5 millamperes:

Type - CW (unmodulated)

Frequency - 60 ±1 MHz

Average power - +20 dBm

(c) Position 3.- With an RF signal having the following characteristics connected to RF INPUT jack (J4), meter shall read 3.0 ±0.5 milliamperes:

Type - CW (unmodulated)

Frequency - 60 ±1 MHz

Average power - +13 dBm

(d) Position 4.- With a gated 60 MHz signal, cosine ^{2} shape, a burst

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repetition rate of 20±0.5KC, amplitude of $1 \begin{smallmatrix} +.5 \\ -0 \end{smallmatrix}$ volt peak-to-peak across a 50 ohm load and a burst width of 280±10 nanoseconds between the 50% amplitude points (see Figure 6), the meter shall read between 5 and 7.5 milliamperes.

(e) Position 5.- The meter shall read between 5 and 7.5 milliamperes when a pulse with the following characteristics is applied to Pulse Input jack J3, and its amplitude is varied from 23.5 to 30 volts.

Amplitude	23.5 to 30.0 volts
Pulse Width	0.5 ±0.1 microseconds
Rep Rate	20 ±0.5 kHz
Distortion	Overshoot/Undershoot less than 0.45 volts

(f) Position 6.- The meter shall read between 5 and 7.5 milliamperes when a pulse with the following characteristics is applied to Pulse Input jack J3, and its amplitude is varied from 13.0 to 18.0 volts.

Amplitude	13.0 to 18.0 volts
Pulse Width	0.5 ±0.1 microsecond
Rep Rate	20 ±0.5 kHz
Distortion	Overshoot/Undershoot less than 0.45 volt

(g) Position 7.- With an RF signal obtained from a 7.5 MHz clock generator, Drawing D10548613 and a 1/4 watt 200 ohm ±5% load connected to the RF INPUT jack (J4), the meter shall read between 5.0 and 8.0 milliamperes.

(h) Position 8.- With an 0.5 microsecond 20 KHz pulse with a ground state of +4 ± 1v and a pulsed state of 0 ± 1v connected to Pulse Input jact (J3), the meter shall read between 5 and 7.5 milliamperes.

(i) Position 9.- With PULSE INPUT jack (J3) shorted to ground, the meter shall read greater than 8.25 milliamperes.

(j) Position 10.- With a + 3.5 volt, 0.5 microsecond, 20 KHz pulse connected to Pulse Input jack (J3) the meter shall read between 5 and 7.5 milliamperes.

3.5.3 High voltage load section.- (See 6.3)

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3.5.3.1 High voltage breakdown and corna.- With +24 volts dc applied to PWR jack (J6), POWER switch (S5) set to ON, and 2A1A3/2A6 switch (S6) set to 2A1A3, there shall be no indication of breakdown or corona in the area of the high voltage cable assembly when a voltage of -11 kV is applied through the high voltage interlock connector. (Breakdown or corona will be considered to have taken place if the load draws in excess of 5 microamps). Verify that the dummy protective cap is mated with connector J7.

3.5.3.2 -9V load input.- With the same conditions as in 3.5.3.1, except with S6 set to 2A6, TEST LOAD meter (M2) shall read between 3.3 and 4.75 milliamperes when a voltage of -4.5 kV is applied through the high voltage interlock connector. With S6 set to 2A1A3, TEST LOAD meter (M2) shall read between 8.6 and 9.4 milliamperes when a voltage of -900 volts is applied between J7-C and the HV INTLK RET.

3.5.4 Range test section.

3.5.4.1 Range and range rate indicators.

3.5.4.1.1 Normal range and range rate indicators.- With +24 volts dc connected to PWR jack (J6), TEST SELECTOR switch (S7) set to OFF, and RANGE switch (S8) set to NORM, each range and range rate indicator lamp (DS2 through DS20) shall illuminate when a logic level voltage of +3 volts dc is applied to the appropriate terminal of J9 or J12 as follows:

Normal Range Indicator Lamp	Reference Designation	J12 Terminal
5 M	DS2	M
10 M	DS3	P
20 M	DS4	N
40 M	DS5	S
80 M	DS6	f
160 M	DS7	g
320 M	DS8	h
640 M	DS9	G
1280 M	DS	H
2560 M	DS11	c

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Range Rate Indicator Lamp	Reference Designation	J9 Terminal
SIGN	DS12	H
SIGN	DS13	G
160 M/S	DS14	M
80 M/S	DS15	N
40 M/S	DS16	P
20 M/S	DS17	S
10 M/S	DS18	f
5 M/S	DS19	g
2.5 M/S	DS20	h

3.5.4.1.2 5000 meter range.- With +3 volts dc connected to J12 e (+3V) and m (+3V RET) and RANGE switch (S8) set to 5000 M only Range indicators DS5, DS7 through DS11 shall illuminate.

3.5.4.2 TEST SELECTOR switch and meter.- The TEST SELECTOR switch (S7) shall be set in turn to positions 1 through 9 with the appropriate test input signal(s) applied for each position. The proper meter indication for each position shall be observed on TEST INDICATOR meter (M3). All dc returns for positions 1 through 8 are corrected to TP17.

(a) Position 1.- The voltage applied to J10-14 required to cause the meter to read left full scale (-310) shall be within the range -9.5 to -10.9 volts dc.

(b) Position 2.- The voltage applied to J10-16 required to cause the meter to read right full scale (+310) shall be within the range +9.5 to +10.9 volts dc.

(c) Position 3.- The meter readings in column 1 shall be produced by a voltage within the corresponding ranges listed in column 2 applied to J10-41.

Column 1	Column 2
Right full scale (+310)	+9.5 to +10.9
Center scale green zone (0)	0
Left full scale (-310)	-9.5 to -10.9

(d) Position 4.- With +10 volts dc connected to J10-16, the voltage applied to J11-39, required to cause the meter to read right full scale (+310) shall be within the range +19.5 to +20.9 volts dc.

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(e) Position 5.- With +10 volts dc applied to J10-16, the voltage applied to J11-41 required to cause a right full scale (+310) meter reading shall be within the range +19.5 to +20.9 volts dc. +10 volts dc applied to both J10-16 and J11-41 shall cause a center scale green zone (0) reading. With 0 volts dc applied to J11-41, the voltage applied to J10-16 required to cause a left full scale (-310) reading shall be within the range +9.5 to +10.9 volts dc.

(f) Position 6.- With +20 volts dc connected to J11-39 and +20 volts dc connected to J11-41, meter shall read at center scale green zone (0).

(g) Position 7.- With +10 volts dc connected to J10-16 and +10 volts dc connected to J10-41, meter shall read at center scale green zone (0).

(h) Position 8.- With -10 volts dc connected to J10-14 and -10 volts dc connected to J10-41, meter shall read at center scale green zone (0).

(i) Position 9.- With return connected to J10-40 and 0 volts dc connected to J10-41, meter shall read at center scale green zone (0).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of one OMTS manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

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4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot information, and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I and II herein. Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE I. - CLASSIFICATION OF DEFECTS

100% INSPECTION

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101.	Continuity	3.4.1	4.6.2.1
102.	Resistance	3.4.2	4.6.2.2
103.	Power switch and indicator lamp	3.4.3	4.6.2.3
104.	Power circuit breaker	3.4.4	4.6.2.4
105.	Power regulation	3.4.5	4.6.2.5

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106.	Synchro differential transmitter alignment	3.5.1.1	4.6.3.1.1
107.	Synchro differential transmitter voltages	3.5.1.2	4.6.3.1.2
108.	Signal selector switch and meter	3.5.2.1	4.6.3.2.1
109.	High voltage breakdown and corona	3.5.3.1	4.6.3.3.1
110.	-9 kV load input	3.5.3.2	4.6.3.3.2
111.	Range and range rate indicators	3.5.4.1	4.6.3.4.1
112.	Test selector switch and meter	3.5.4.2	4.6.3.4.2

4.3.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.- One OMTS out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table II.

TABLE II. - SPECIAL SAMPLING

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable dwg. visual
302.	General specification	3.2	MIL-F-13926, MIL-C-46357, visual
303.	Vibration	3.3.1	4.6.1.1
304.	Shock	3.3.2	4.6.1.2
305.	Altitude - operating	3.3.3.1	4.6.1.3
306.	Altitude - nonoperating	3.3.3.2	4.6.1.3

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307.	Temperature	3.3.4	4.6.1.4
308.	Humidity - nonoperating	3.3.5	4.6.1.5
309.	Drop test	3.3.6	4.6.1.6
310.	Pressure retention	3.3.7	4.6.1.7
311.	Sand and dust	3.3.9	4.6.1.9
312.	Salt spray	3.3.10	4.6.1.10

4.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

4.5. Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government-furnished inspection equipment.- Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Contractor design.- The contractor shall design and supply inspection equipment compatible with "Test Methods and Procedures" specified in 4.6 of this specification and with the "Component Inspection" procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is considered normally to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to ensure that the test equipment will indicate reliable acceptability of a product that does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of the test equipment.

4.5.2.2 Test equipment.- In conjunction with 4.5.2.1, the following test equipment shall be used in the performance of the applicable tests as specified in 4.6:

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Item Number		Description
1.	Ohmmeter	To measure up to 10.0 megohms ±2%.
2.	Power supply	0 to 30 volts dc ±5%, 0 to 2 amperes, regulated, less than 1% ripple.
3.	Voltmeter, ac-dc	Ranges of 0 to 200 volts ac, 0 to 30 volts dc, accuracies of ±2% on dc and ±3% on ac, sensitivities 20,000 ohms per volt dc and 5,000 ohms per volt ac.
4.	Synchro control transmitter (CX)	115 volts, 400 Hz (11CX4e-M20708/2-001).
5.	Ac power supply	115 volts rms ±5%, 400 Hz ±5%.
6.	Oscilloscope	Rise time of scope plus associated probe, 23 nanoseconds or less; input impedance 1 megohm or greater, paralleled by input capacitance of 47 picofarads; sweep range, 0.01 microsecond/cm; sensitivity, 50 millivolts/cm.
7.	Oscillator	60 MHz Local Oscillator, D10548158; frequency 59.936 MHz ±1000 Hz; output power 200 mW minimum into 50 ± 5 ohms; power input 28 ± 1Vdc not to exceed 3 watts.
8.	Pulse generator	Repetition rates of 250 Hz to 20 kHz. Rise time less than 12 nanoseconds for positive pulse, less than 10 nanosceonds for negative pulse. amplitude 0 to 30 volts into a 100-ohm load.
9.	7.5 MHz clock generator	Drawing D10548613, assembly 4A1 of Radar Set AN/VPS-2.
10.	Power supply	5 volts dc ±1%, 0.5 ampere, regulated, less than 1% ripple.
11.	Power supply (high voltage)	0 to -20 kV, 10 mA dc, unregulated.
12.	Dc power supply	0 to 30 volts dc, ±5%, 0.5 ampere.
13.	Dc power supply	0 to 20 volts dc, ±5%, 0.5 ampere.

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14.	Capacitive load	1,000 pF, ±5%.
15.	Connector	BNC tee
16.	DC Power Supply	+15 ± 1Vdc regulated; ac ripple less than 50 mV peak-to-peak; capacity 100mA
17.	DC Power Supply	-15 ± 1Vdc regulated; ac ripple less than 50 mV peak-to-peak; capacity 100mA
18.	Hybrid Junction	10548183
19.	Transmit Gate	10549202
20.	Variable Attenuator	10558457
21.	Electrical Plug Connector	10556252
22.	Resistor	200 ohm, 1/4 watt, ±5%
23.	Pulse Generator	0 to ±5v, PRF of 20 KHz pulse width of 0.5 microsecond, variable DC ground state of 0 to + 5 volts.
24.	Cable R.F.	10554554

4.6 Test methods and procedures.

4.6.1 Environmental.- Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.6.1.1 Vibration.- The test shall be performed as specified in Method 514, Procedure IX, Part I, of MIL-STD-810 except that Failure criteria of 4.9.1.3 does not apply. After vibration, the OMTS shall be tested as given in the subparagraphs of 4.6.2 and 4.6.3 for compliance with 3.3.1.

4.6.1.2 Shock.- This test shall be performed as specified in Procedure I of Method 516 of MIL-STD-810 except that shock pulses be 11.0 ms and 15 g's. The g value of each impulse shall be within ±10 percent when measured with a 0.2 to 250 Hz filter and the maximum g shall occur at approximately 5-1/2 milliseconds. After the shock impulses, the OMTS shall be tested as specified in 4.6.2 and 4.6.3 for compliance with 3.3.2.

4.6.1.3 Altitude - non-operating and operating.- This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in Step 2 the altitude shall be limited to 40,000 ft. (5.5 in Hg). Operation of

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Step 3 shall be the performance tests of 4.6.3. This test shall be performed with the case closed and latched.

4.6.1.4 Temperature.

4.6.1.4.1 High (operating and non-operating).- This test shall be performed as specified in Procedure I, Method 501, of MIL-STD-810. The temperature of Step 2 shall be +185°F. The temperature of Step 4 shall be +130°F. In Step 5 the OMTS shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.4.2 Low (operating and non-operating).- This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810. The temperature of Step 2 shall be -85°F maintained for 24 hours. The temperature of Step 4 shall be -40°F. In Steps 5 and 7, the OMTS shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.5 Humidity-nonoperating.- This test shall be performed in accordance with Procedure I, Method 507 or MIL-STD-810 except that in Step 4, the relative humidity shall be maintained at 95%. This test shall be performed with the case closed and latched.

4.6.1.6 Drop test.- This test shall be performed as specified in Procedure II, Method 516 of MIL-STD-810 with conditions as specified in 3.3.6. After completion of the test, the case shall be inspected for compliance with 3.3.6.

4.6.1.7 Pressure retention.- The OMTS breather valve shall be removed and replaced by an appropriate plug. A suitable connection shall be made for the introduction of air pressure. A gauge shall be connected to indicate pressure within the container. Air pressure shall be applied up to 1 pound per square inch and the intake valve closed to retain pressure. A soap solution shall be applied to all joints, seams, and welds. There shall be no evidence of leakage. After 5 minutes, the gauge shall be observed to make sure that original pressure has been maintained. This test shall be performed with the case closed and latched.

4.6.1.8 Electromagnetic Interference.- One of the first 50MTS produced shall be tested for interference in accordance with MIL-STD-462. After approval, subsequent acceptance for electromagnetic interference requirements may be by certification.

4.6.1.9 Sand and dust.- This test shall be performed in accordance with Procedure I, Method 510 of MIL-STD-810 except that humidity shall be maintained between 15 and 30% at a temperature of plus 75°F, air velocity shall be maintained at a minimum of 15 knots, particle size and density shall be as stated in 3.3.9 and length of test shall be 3 hours. This test shall be performed with the case closed and latched. After completion of the test, the OMTS shall be tested for compliance with 3.4 and 3.5.

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4.6.1.10 Salt spray.- This test shall be performed in accordance with Procedure I, Method 509 of MIL-STD-810 with the case closed and latched. After completion of the test, the OMTS shall be tested for compliance with 3.4 and 3.5.

4.6.1.11 Fungus.- One of the first 5 OMTS produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.6.2 Electrical.

4.6.2.1 Continuity.- Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.1.

4.6.2.2 Resistance.- Resistance shall be measured with an ohmmeter, item 1, for compliance with 3.4.2.

4.6.2.3 Power switch and indicator lamp.- Connect +24 volts dc power from variable power supply, item 2, to the OMTS with connections made as shown in Figure 1. With 24V PWR circuit breaker (CB1) energized, POWER indicator lamp (DS1) shall extinguish when POWER switch (S5) is set to OFF. Operate POWER switch to ON-OFF three times. Adjust power supply over the range of +22 to +30 volts dc. With POWER switch set to ON, POWER indicator shall remain illuminated.

4.6.2.4 Power circuit breaker.- Energize 24V PWR circuit breaker (CB1) by depressing reset pushbutton so white band is not exposed. Connect dc power as in 4.6.2.3. Set POWER switch (S5) to ON and observe that POWER indicator lamp (DS1) is illuminated. Deenergize circuit breaker by pulling out reset pushbutton so that white band is exposed and observe that POWER indicator lamp is extenguished. Turn off dc power, reset circuit breaker, and turn on dc power. POWER indicator lamp shall illuminate.

4.6.2.5 Dc power regulation.- With the OMTS panel removed from the case, energize 24V PWR circuit breaker (CB1), connect dc power as in 4.6.2.3 and set POWER switch (S5) to ON. Connect voltmeter, item 3, across diode VR1. Adjust power supply over the range of +22 to +30 volts dc. Voltage across CR7 shall be 4.85 to 5.35.

4.6.3 Performance.

4.6.3.1 Antenna test section.

4.6.3.1.1 EL/TRAV POS synchro differential transmitter alignment.- Connect external synchro control transmitter (CX), item 4 aligned and locked at electrical zero and excited by 115 volts, 400 Hz, from ac power supply, item 5, to the EL/TRAV POS synchro differential transmitter (B1) of the OMTS in accordance with Figure 2.

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(a) Connect oscilloscope, item 6, across S1-S3 of EL/TRAV POS synchro differential transmitter B1. A reading less than 600 millivolts peak-to-peak (null) shall be observed on the oscilloscope to confirm electrical zero input to B1.

(b) Turn off ac power to the external CX. Set the EL/TRAV POS dial on the OMTS to 0°. Place a jumper across S1-S3 of B1. Connect oscilloscope across R1-R3 of B1. Turn on ac power to external CX. A reading less than 600 millivolts peak-to-peak shall be observed on the oscilloscope. (This is a "null" reading, which indicates that B1 is aligned either at 0° or 180°).

(c) Turn off ac power to external CX. Place a jumper across S1-R1 of B1. Connect voltmeter, item 3, across S2-R2 of B1. Turn on ac power to external CX. A reading of 0 to 12 volts rms (nominal) shall be observed on volmeter indicating that B1 is aligned at 0° in compliance with 3.5.1.1. (If P1 is aligned at 180°, a nominal voltage reading of 160 ±12 volts rms will be observed.)

(d) Remove all jumper wires.

4.6.3.1.2 EL/TRAV POS synchro differential transmitter voltages.- Connect external synchro control transmitter (CX), item 4, aligned and locked at electrical zero and excited by 115 volts, 400 Hz, from ac power supply, item 5, to EL/TRAV POS synchro differential transmitter (B1) of the OMTS in accordance with Figure 2.

(a) Ac voltages as specified in 3.5.1.2 shall be observed.

(b) Connect voltmeter across R1-R2 of B1. Repeat step (a).

(c) Connect voltmeter across R2-R3 of B1. Repeat step (a).

4.6.3.2 Signal test section.

4.6.3.2.1 Signal selector switch and meter.- Connect +24 volts dc input power from power supply, item 2, to the OMTS in accordance with Figure 1. SIGNAL SELECTOR switch (S4) shall be operated in positions 1 through 9 and indications shall be observed on SIGNAL INDICATOR meter (M1) in accordance with the following procedures:

Note. Both maximum deflection and zero deflection of the meter are indications that there is either no signal input to the Signal Test Section or that a malfunction exists when a signal is present.

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(a) Position 1.- Set SIGNAL SELECTOR switch to Position 1. Connect oscillator, item 7, to RF INPUT jack (J4). The oscillator signal shall be as specified in 3.5.2.1, Position 1. Meter shall read between 5 and 7.5 milliamperes. See Figure 4.

(b) Position 2.- Set SIGNAL SELECTOR switch to Position 2. Connect oscillator, item 7, RF INPUT jack (J4). The oscillator signal shall be as specified in 3.5.2.1, Position 2. Meter shall read between 5 and 7.5 milliamperes. See Figure 3 for Position 2 through Position 9.

(c) Position 3.- Set SIGNAL SELECTOR switch to Position 3. Connect oscillator, item 7, to RF INPUT jack (J4). The oscillator signal shall be as specified in 3.5.2.1, Position 3. Meter shall read 3.0 ±0.5 milliamperes

(d) Position 4.- Set SIGNAL SELECTOR switch to Position 4. Connect test setup as shown in Figure 3 with Item 24 connected to J4 of OMTS, with pulse characteristics as specified in 3.5.2.1 (d). The meter shall read between 5 and 7.5 milliamperes. Prior to next test disconnect 50 OHM load.

(e) Position 5.- Set SIGNAL SELECTOR switch to Position 5. Attach BNC Tee, item 15, and capacitive load, item 14, to output of pulse generator, item 8; then set pulse generator as specified in 3.5.2.1, Position 5. Connect pulse generator to PULSE INPUT jack (J3). Meter shall read between 5 and 7.5 milliamperes.

(f) Position 6.- Set SIGNAL SELECTOR switch to Position 6. Attach BNC Tee, item 15, and capacitive load, item 14, to output of pulse generator, item 8; then set pulse generator as specified in 3.5.2.1, Position 6. Connect pulse generator to PULSE INPUT jack (J3). Meter shall read between 5 and 7.5 milliamperes.

(g) Position 7.- Set SIGNAL SELECTOR switch to Position 7. Using oscillator, item 7, 7.5 MHz clock generator, item 9, and +5 volts dc power supply, item 10. The input signal to the OMTS shall be as specified in 3.5.2.1, Position 7. Meter shall read between 5 and 8.0 milliamperes.

Caution.- Do not select Position 8 until proper signal level of the input has been set.

(h) Position 8.- Connect pulse generator (Item 23) to Pulse Input jack (J3). Set the PRF, pulse width, DC bias and amplitude as specified in 3.5.3.1 (h). Set signal selector to position 8. Meter shall read between 5 and 7.5 milliamperes as specified.

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(i) Position 9.- Set SIGNAL SELECTOR switch to Position9, Short PULSE INPUT jack (J3) to ground as specified in 3.5.2.1, Position 9. Meter shall read greater than 8.25 milliamperes.

(j) Position 10.- Connect pulse generator, Item 8 to Pulse Input (J3). Adjust output of pulse generator for a +3.5 volt, 0.5 microsecond, 20 KHz pulse as specified in 3.5.2.1 (j). Set signal selector switch to position 10. Meter shall read between 5 and 7.5 milliamperes.

4.6.3.3 High voltage load section.

WARNING. THE FOLLOWING PROCEDURES REQUIRE THE APPLICATION OF VOLTAGES THAT ARE DANGEROUS TO LIFE. THESE PROCEDURES SHALL NOT BE PERFORMED UNTIL AFTER THE PERFORMANCE OF RESISTANCE MEASUREMENTS SPECIFIED IN 4.6.2.2 IN COMPLIANCE WITH THE REQUIREMENTS OF 3.4.2.2. DEATH ON CONTACT MAY RESULT FROM FAILURE TO OBSERVE SAFETY PRECAUTIONS.

4.6.3.3.1 High voltage breakdown and corona.- Connect high voltage power supply, item 11, to the OMTS in accordance with Figure 5. Connect +24 volts dc to 24V PWD jack (J6) in accordance with Figure 1. Set POWER switch (S5) to ON. Set 2A1A3/2A6 switch (S6) to 2A1A3 position. Increase power supply output from 0 to -11 kV slowly. There shall be no indication of breakdown or corona in compliance with 3.5.3.1. Decrease power supply output to 0 volt. Breakdown or corona will be considered to have taken place if the load draws in excess of 5 microamps.

4.6.3.3.2 -9kV load input.- With the same test conditions as 4.6.3.3.1, set 2A1A3/2A6 switch (S6) to 2A6 position. Increase output of power supply slowly from 0 to -4.5 kV. At -4.5 kV, TEST LOAD meter (M2) shall indicate between 3.6 and 4.75 milliamperes in compliance with 3.5.3.2. Set 2A1A3/2A6 switch (S6) to 2A1A3 position. Leave return in same place as in 4.6.3.3.1. Connect -900V to J7-C. TEST LOAD meter (M2) shall indicate between 8.6 and 9.4 milliamperes in compliance with 3.5.3.2.

4.6.3.4 Range test section.

4.6.3.4.1 Range and range rate indicators.- Connect +24 volts dc input power to the OMTS in accordance with Figure 1. Set TEST SELECTOR switch (S7) to OFF and RANGE switch (S8) to NORM. Using power supply, item 10, apply +3 volts dc to appropriate terminals of J12 and J9 as specified in 3.5.4.1 and observe that corresponding lamp is illuminated.

4.6.3.4.2 TEST SELECTOR switch and meter.- TEST SELECTOR switch (S7) shall be observed on TEST INDICATOR meter (M3) in accordance with the following procedures. Dc input voltages shall be obtained from power supplies, item 12 and 13. Dc returns shall be connected to TP17 for positions 1 through 8.

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(a) Position 1.- Set TEST SELECTOR switch to Position 1. Connect power supply to J10-14. Vary voltage in the range -9.5 to -10.9 volts dc until meter reads left full scale (-310).

(b) Position 2.- Set TEST SELECTOR switch to Position 2. Connect power supply to J10-16. Vary voltage in the range +9.5 to +10.9 volts dc until meter reads right full scale (+310).

(c) Position 3.- Set TEST SELECTOR switch to Position 3. Connect power supply to J10-41. Vary voltage in the range +9.5 to +10.9 volts dc until meter reads right full scale (+310). Reduce the voltage to 0 volts; the meter shall read in the center scale green zone (0). Vary voltage in the range -9.5 to -10.9 volts dc until meter reads left full scale (-310).

(d) Position 4.- Set TEST SELECTOR switch to Position 4. Connect +10 volts dc to J10-16. Connect other power supply to J11-39. Vary voltage in the range +19.5 to +20.9 volts dc until meter reads right full scale (+310).

(e) Position 5.- Set TEST SELECTOR switch to Position 5. Connect +10 volts dc to J10-16. Connect other power supply to J11-41 and vary the voltage in the range +19.5 to +20.9 volts dc until meter reads right full scale (+310). Change voltage on J11-41 to +10 volts; meter shall read center scale green zone (0). Reduce the voltage on J11-41 to 0. Vary the voltage on J10-16 in the range +9.5 to +10.9 volts until meter reads left full scale (-310).

(f) Position 6.- Set TEST SELECTOR switch to Position 6. Connect +20 volts dc to J11-39 and +20 volts dc to J11-41. Meter shall read at center scale green zone (0).

(g) Position 7.- Set TEST SELECTOR switch to Position 7. Connect +10 volts dc to J10-16 and +10 volts dc to J10-41. Meter shall read at center scale green zone (0).

(h) Position 8.- Set TEST SELECTOR switch to Position 8. Connect -10 volts dc to J10-14 and -10 volts dc to J10-41. Meter shall read at center scale green zone (0).

(i) Position 9.- Set TEST SELECTOR switch to Position 9. Connect return to J10-40 and 0 volts dc to J10-41. Meter shall read at center scale green zone (0).

5. PREPARATION FOR DELIVERY

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5.1 Packaging.- Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10554522. The level of protection shall be as specified in the contract.

6. NOTES

6.1 Intended use.- The OMTS is a portable test set for use by an organizational maintenance technician (second echelon) to locate defective units, assemblies, and subassemblies of Radar Set AN/VPS-2. The OMTS contains four test sections assembled on a common panel and mounted in a protective combination case: (1) The antenna test section isolates problems between the servo electronics assemblies and the elevation and traverse drive assemblies. (2) The signal test section detects the presence and absence of RF and pulsed signals. (3) The high voltage load section provides a load to test and isolate malfunction between the klystron beam power supply assembly, klystron power supply assembly, and the klystron power amplifier subassembly. (4) The range test section provides information to determine whether the range computer is free of catastrophic failure and to isolate a failure to within a group of four range computer circuit boards.

6.2 Ordering data.- Procurement documents should specify the following.

(a) Title, number and date of this specification

(b) Applicable packaging data sheet (see 5.1)

(c) Selection of applicable level of preservation packaging and packing

(d) Level of protection required

(e) Applicable stock number

(f) Block of serial numbers to identify each OMTS

6.3 High voltage warning.- High voltages dangerous to life must be applied to the OMTS in performing tests on the high voltage load section as specified in 4.6.3.3. Death on contact may result from failure to observe safety precautions. Prior to applying high voltage, the resistance measurements specified in 4.6.2.2 should be performed in compliance with the requirements of 3.4.2.2.

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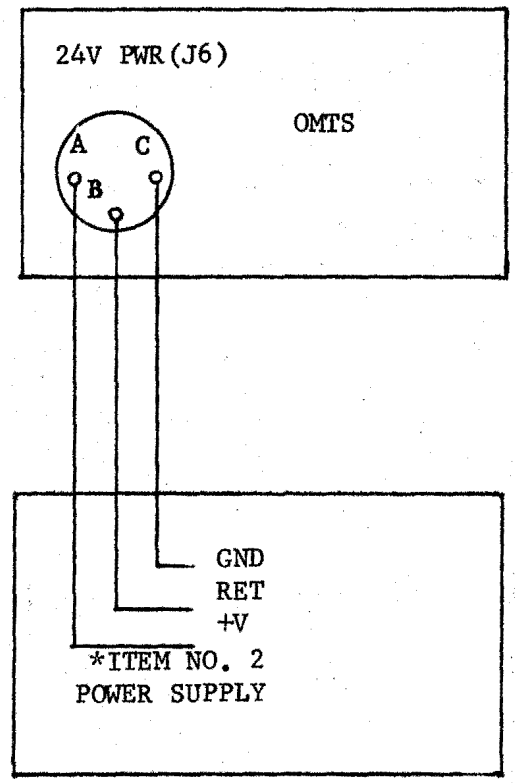
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Preparing activity

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*REFER TO 4.5.2.2

FIGURE 1. POWER INPUT TEST SETUP

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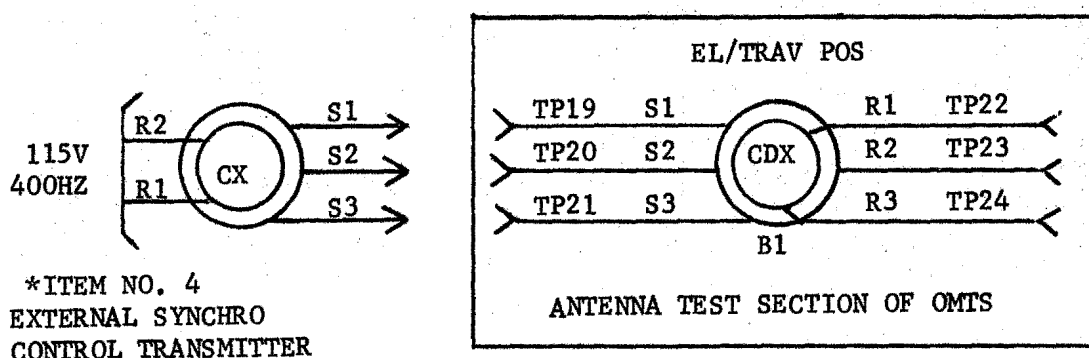


FIGURE 2. SYNCHRO INPUT TEST SETUP

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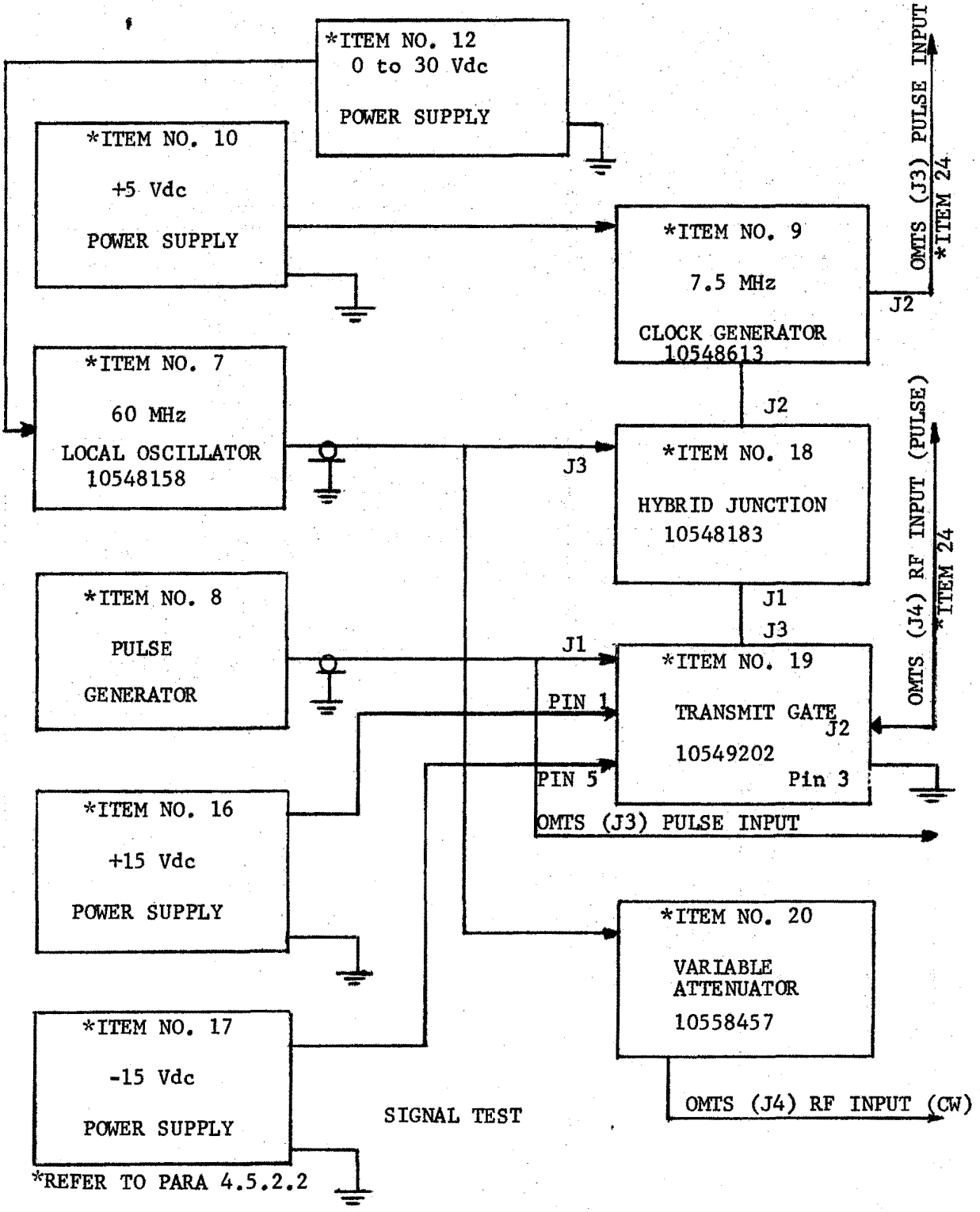
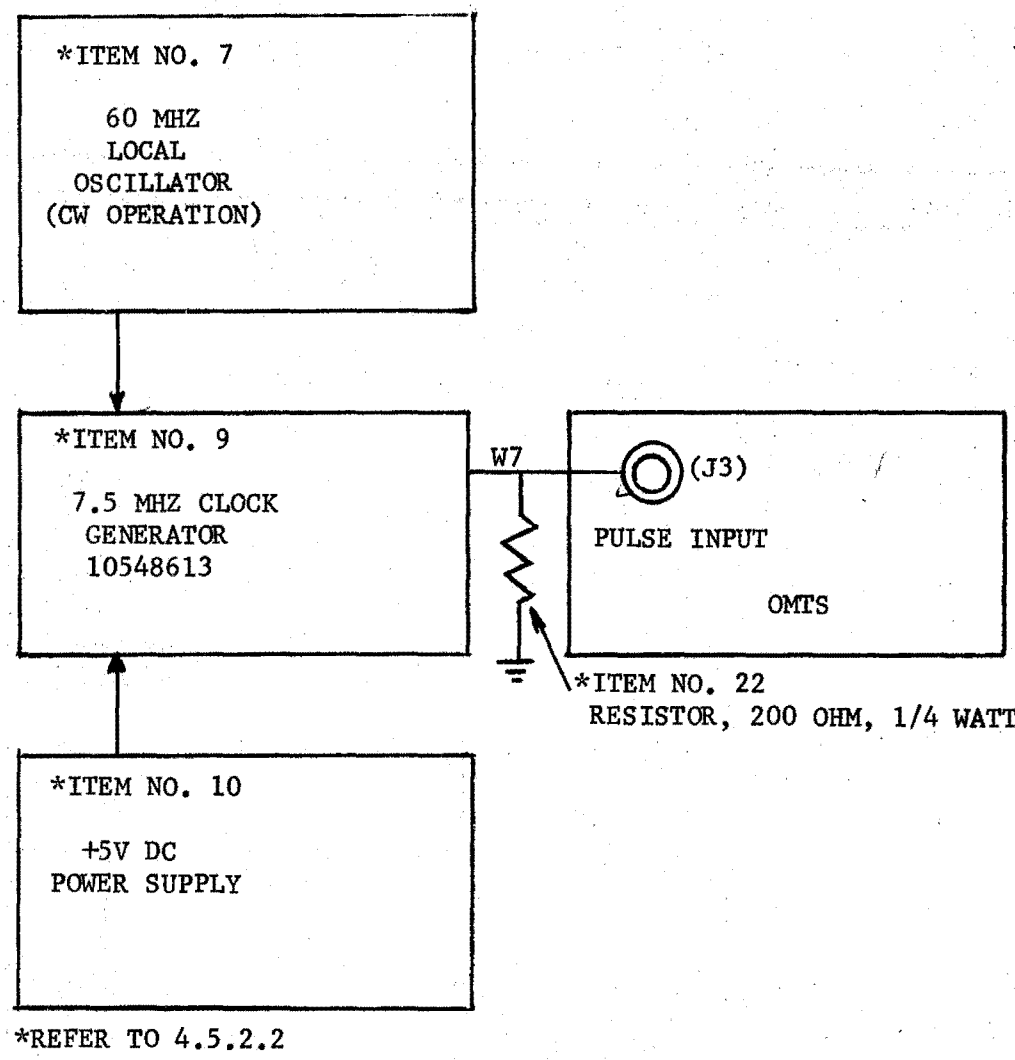


FIGURE 3

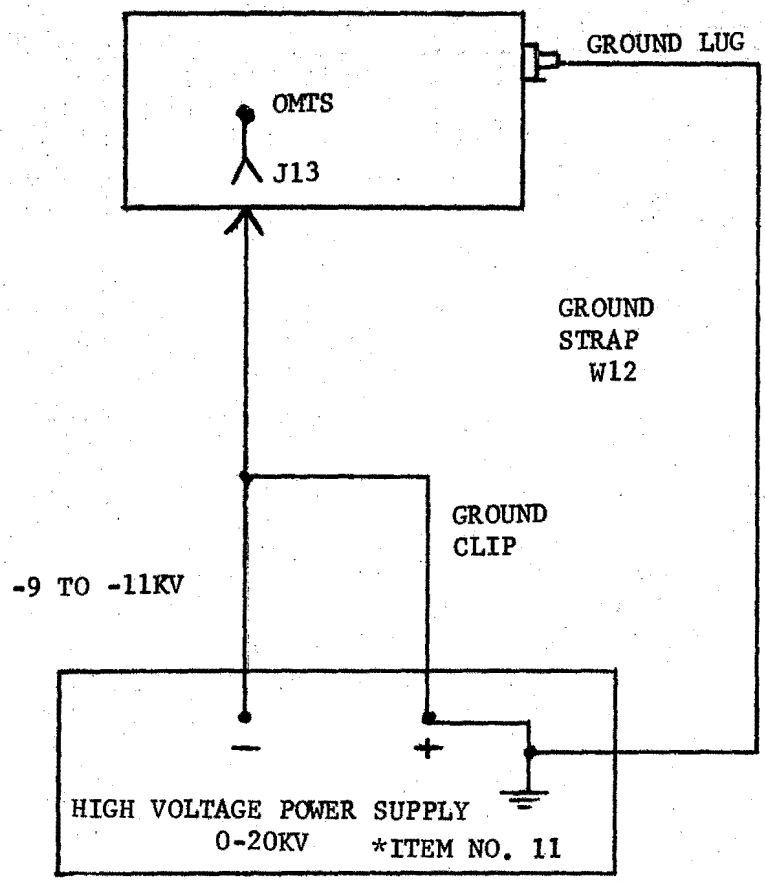
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7.5 MHZ CLOCK PULSE
INPUT TEST SETUP

FIGURE 4

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*REFER TO 4.5.2.2

HIGH VOLTAGE LOAD
TEST SETUP

FIGURE 5

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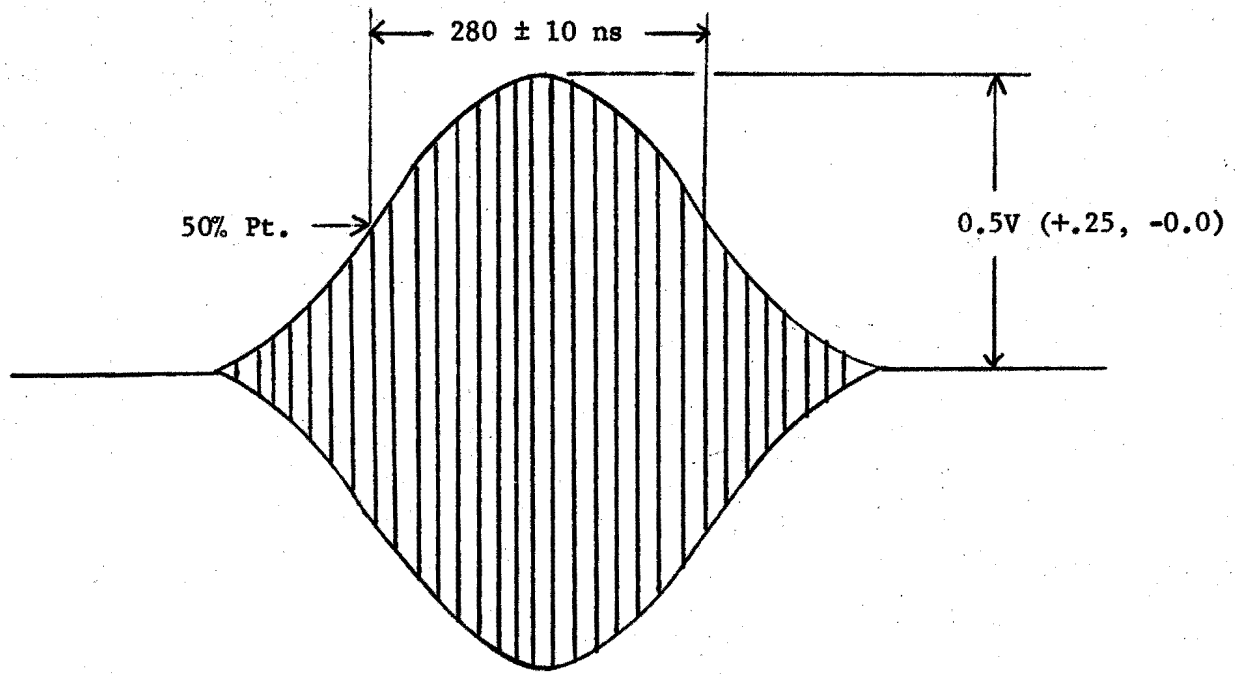


FIGURE 6. Rf BURST.

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